

Macroevolutionary Rates of Species Interactions: Approximate Bayesian Inference from Cophylogenies

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Abstract

Understanding the macroevolutionary dynamics of species interactions such as parasitisms, commensalisms, and mutualisms is an important goal in evolutionary ecology. To this end, statistical inference from time-calibrated cophylogenies holds promising potential. However, such inference cannot yet quantify the rates of different types of speciation and extinction that occur in the host and symbiont clades on the same timeline. Here we present an Approximate Bayesian Computation (ABC) approach that infers rates of six types of speciation or extinction from a cophylogenetic system: (i) host speciation, (ii) symbiont speciation without host-switching, (iii) symbiont speciation with host-switching, (iv) cospeciation, (v) host extinction, and (vi) symbiont extinction. The ABC approach relies on a novel design of summary statistics based on the density curves of pairwise Branch Length Differences (BLenD) of the cophylogeny, which are informative about the relative relationships (ratios) between the six speciation/extinction rates in a single cophylogeny. Here we describe two levels of inference that differ in informativeness and data requirement: the first level infers speciation/extinction rates relative to the total net diversification rate of the cophylogeny without needing information on the time frame of the cophylogeny; the second level infers absolute speciation/extinction rates in units of events per lineage per million years using information on the time frame of the cophylogeny. When the target cophylogeny is sufficiently large, both levels of inference achieve clearly improved accuracy relative to the prior, and both are reasonably honest about uncertainty. Using a cophylogenetic dataset of Batesian mimicry of *Pachyrhynchus* by *Doliops* weevils, we show (1) that the first level of inference is sufficient to quantify the relative rates of different speciation/extinction processes *within* the target cophylogeny and (2) that the second level of inference allows potentially comparing speciation/extinction rates in the target cophylogeny to those in another cophylogeny (i.e., *cross-cophylogeny* comparisons). We discuss potential improvements for the use of the BLenD curves as summary statistics for simulation-based inference, including potential applications in machine learning approaches. Understanding speciation and extinction rate variation within and between cophylogenetic systems, enabled by this approach and an increasing availability of time-calibrated cophylogenies, has potential implications for various areas in ecology and evolution such as host conservatism, trait-driven diversification, and pathogen spillover risk.

Key words: *Phylogenetic comparative methods, Likelihood-free inference, Birth-death processes, Diversification rates, Coevolution, Co-diversification, Symbiosis*

Introduction

Biologists have long sought to understand the relationship between macroevolution and ecological interactions. Early examples include how Darwin explains the diversity of life as a result of natural selection in his seminal book, using examples where natural selection occurs as a result of organisms' "struggle for life" against their competitors and natural enemies (Darwin 1859). A recent comprehensive synthesis has consolidated the macroevolution of species interactions as a rapidly evolving subfield of ecology and evolution with deep historical roots (see Hembry and Weber 2020 for an overview). Several other syntheses have separately examined the effects of species interactions on macroevolution, reaching varying conclusions about whether generalities exist in how species interactions influence clade diversification (Jablonski 2008; Harmon et al. 2019; Zeng and Wiens 2021; Kaur and Pennell 2023). Another body of research has focused on the effects of macroevolution on species interaction networks. Such examples include how macroevolutionary stability shapes the intricate architecture of species interaction networks (Burin et al. 2021) and statistical inference of the evolutionary history of species interaction networks (Braga et al. 2020, 2021). Recent years have also seen significant advances in theoretical work that, by modeling how speciation arises from mutation in a (meta)community with ongoing species interactions, bridges the microevolution-macroevolution gap and provides mechanistic insights into the evolution of ecological communities (Aguilée et al. 2018; Coelho and Rangel 2018; Maliet et al. 2020; Chaparro-Pedraza et al. 2022; Pontarp et al. 2024; Zeng and Hembry 2024).

In particular, the joint macroevolutionary history of organisms that interact in pairs (i.e., bipartite interactions) has attracted much attention over the decades. Examples of bipartite interactions include those between plants and pollinators, plants and seed dispersers, hosts and parasites, and between plants and herbivores (see Bronstein 2015 for an overview). Coevolution can arise when reciprocal evolutionary change happens between interacting parties, driving or impeding the diversification of the interacting clades (Yoder and Nuismer 2010; Hembry et al. 2014). Coevolution can leave detectable signals in the cophylogenetic patterns of bipartite species interactions, such as those formed by hosts and their mutualistic, commensal, or parasitic symbionts. Significant effort has been devoted to studying topological congruence between phylogenies of interacting organisms as an indicator of cospeciation or codiversification (e.g., Legendre et al. 2002; Vienne et al. 2007; Hoyal et al. 2012; Hayward et al. 2021; Perez-Lamarque and Morlon 2024; reviewed in Nieberding and Olivieri 2007; Vienne et al. 2013; Martínez-Aquino 2016; Dismukes et al. 2022). Recent cophylogenetic research has forayed into linking cophylogenetic patterns with their underlying eco-evolutionary processes (Blasco-Costa et al. 2021). Another area of active research is cophylogenetic reconstruction, aiming to reconstruct the joint evolutionary history of interacting clades by finding a parsimonious series of events that give rise to the observed cophylogenetic patterns (e.g.,

Merkle et al. 2010; Baudet et al. 2015; Sinaimeri et al. 2023; reviewed in Charleston and Libeskind-Hadas 2014).

However, the study of the macroevolutionary rates of cophylogenetic systems remains in its infancy. Macroevolutionary rates can be defined, for single clades, as speciation and extinction rates that correspond to the rate parameter in a Poisson-like process (Liow, Quental, and Marshall 2010; Rabosky et al. 2007; Stadler 2013; Harmon et al. 2014). Here, we define macroevolutionary rates of species interactions (i.e., of interacting clades) as speciation and extinction rates that can be inferred from cophylogenetic systems. Although much recent work has been focused on the advancement of phylogenetic comparative methods for estimating the speciation, extinction, or net diversification rates of single clades (Garamszegi 2014; Revell and Harmon 2022; Morlon et al. 2024), single-clade-based phylogenetic methods cannot be readily used for cophylogenetic systems. Specifically, although these single-clade-based methods allow diversification rates to be inferred separately for host and symbiont clades in a given cophylogenetic system, these rate estimates cannot provide a full picture of the prevalence of different types of speciation in the cophylogeny (Charleston and Perkins 2006; Charleston and Libeskind-Hadas 2014). Motivated by a desire to better understand parasitic symbionts (e.g., pathogens), Alcala et al. (2017) represents an important advance in estimating macroevolutionary rates of species interactions from cophylogenetic systems. However, this approach has two features that prevent it from providing a full picture of the macroevolutionary events in a cophylogenetic system: (i) the host phylogeny is considered as given and known, so host speciation is not allowed to occur on the same timeline as symbiont speciation and cospeciation; (ii) it does not take into account host extinction, which could leave important signature on cophylogenetic patterns by causing the extinction of the extinct host's associated symbionts or by transferring the extinct host's associated symbionts to other, extant hosts (i.e., "transfer from the dead"; Menet et al. 2022). A promising recent approach can already tease apart the relative contributions of different macroevolutionary processes in the symbiont clades (host switching, pollinator speciation, and pollinator extinction) in generating cophylogenetic patterns, but this approach does not infer host speciation rates (Satler et al. 2019). Fortunately, recent advances have allowed simulating cophylogenetic systems as a result of constant-rate birth-death processes, simultaneously incorporating speciation and extinction in both the host and symbiont clades (Dismukes and Heath 2021). This new model has paved the way for developing new statistical methods for estimating macroevolutionary rates of bipartite species interactions.

Approximate Bayesian Computation (ABC) has been shown to be a useful tool for estimating macroevolutionary rates of species interactions from cophylogenetic systems (Alcala et al. 2017). ABC is a class of simulation-based inference that has proven especially suitable for models whose likelihoods are impossible or difficult to obtain (Marin et al. 2012; Sunnåker et al. 2013). Initially developed by geneticists, ABC has been increasingly used in ecology and evolution (Pantel and Becks 2023). In the most basic form of ABC, real data (the observation) is compared to numerous simulations generated using different parameter samples, and parameter samples that generate simulations similar enough to the real data constitute the approximate posterior distribution of that parameter; the key to successful ABC is designing summary statistics that are informative and low in dimensionality (Sunnåker et al. 2013).

Conceptual frameworks and standard practices have been, and are being, established for using approximate Bayesian computation for both parameter estimation and model selection in ecology and evolution (Csilléry et al. 2010; Csilléry et al. 2012; Janzen et al. 2015; Pontarp et al. 2019; Pantel and Becks 2023). As for estimating macroevolutionary rates of species interactions from cophylogenetic systems, the existing ABC approach is promising, but it is limited in the number of inferable rate parameters.

Here, we develop a new ABC approach that infers the rates of six types of speciation or extinction from an extant cophylogeny: (i) host speciation, (ii) symbiont speciation without host-switching, (iii) symbiont speciation with host-switching, (iv) cospeciation, (v) host extinction, and (vi) symbiont extinction. This ABC approach relies on newly designed summary statistics based on density curves of pairwise Branch Length Differences (BLenD) and the sizes of the host and symbiont trees. We show that the ABC can achieve clearly improved accuracy relative to the prior with reasonable honesty about uncertainty when the target cophylogeny is sufficiently large. The empirical application of the ABC approach is demonstrated with an example.

Methods

In brief, we follow these two steps to estimate the speciation and extinction rates of a target cophylogeny, defined as a system consisting of two time-calibrated phylogenetic trees and the network formed by the tips of the two trees: (i) We simulate cophylogenies from a prior distribution of speciation and extinction rates following *a birth-death model for co-diversification*. (ii) In an *ABC framework*, the simulated cophylogenies are compared to the target cophylogeny to obtain the posterior distribution of speciation and extinction rates. Notably, the similarity between the simulated and target cophylogenies is decided by *summary statistics*. In our *performance evaluation*, we test the ABC procedure against a test set of cophylogenies whose speciation and extinction rates are drawn from the prior distribution. In our *application to empirical data*, we perform the ABC procedure using an empirical cophylogeny as the target cophylogeny. Below, we provide details of each of these components.

A Birth-Death Model for Co-diversification

We use the `sim_cophyBD` function from the R package `treeduck` (ver. 1.1.0; Dismukes and Heath 2021) for simulating cophylogenies following a birth–death model. The model in `treeduck` allows the simulation of host and symbiont co-diversification (Fig. 1). Beginning with both the host and symbiont clades having one lineage, the simulator considers six types of speciation or extinction events (Fig. 2): host speciation (λ_H), symbiont speciation without host switching (λ_S), cospeciation (λ_C), symbiont speciation with host switching (λ_W ; denoted by in χ

in Dismukes and Heath 2021), host extinction (μ_H), and symbiont extinction (μ_S). Each type of event is controlled by a rate parameter that represents the number of events per lineage per unit time.

In extant cophylogenetic systems, the extinction rate must not exceed the speciation rate for the host and symbiont clades. We further parameterize the model by defining relative extinction rates ϵ_H (for the host clade) and ϵ_S (for the symbiont clade) as follows,

$$\epsilon_H = \frac{\mu_H}{\lambda_H + \lambda_C}, \quad \epsilon_S = \frac{\mu_S}{\lambda_S + \lambda_C + \lambda_W}$$

where both ϵ_H and ϵ_S range between 0 and 1.

ABC Framework

Parameters of interest can be collectively seen as a multivariate parameter θ , which is treated as a random variable in Bayesian parameter inference. The posterior probability of θ given a target cophylogeny T can be, in theory, obtained by

$$P(\theta|T) = \frac{P(T|\theta)P(\theta)}{\int P(T|\theta)P(\theta)d\theta}$$

where $P(\theta)$ is known as the prior distribution and $P(T|\theta)$ as the likelihood function (i.e., the probability of observing the cophylogeny T given the parameter θ). However, since T is a cophylogeny and θ is the parameter in the birth-death model, the likelihood function of the cophylogenetic birth-death model (Dismukes & Heath 2021) is difficult to obtain, making Approximate Bayesian Computation (ABC) an especially useful approach for parameter estimation (Alcala et al. 2017). An ABC approach does not explicitly compute the likelihood but instead uses simulations to approximate the posterior distribution. Here, we use the basic form of a rejection-based ABC approach (Csilléry et al. 2012) following these steps. First, cophylogenies are simulated with the birth-death model (see Simulating Cophylogenies from the Prior Distribution in Methods) using samples of θ drawn from the prior distribution. Second, distances are computed between each simulated cophylogeny and the target cophylogeny T (see Distance Between Target and Simulated Cophylogenies in Methods). Third, all simulated cophylogenies are ranked from the least to the most distant from D . Fourth, a tolerance rate (also known as acceptance rate) decides what proportion of simulated cophylogenies with the smallest distance to D are accepted. Eventually, values of θ of the accepted cophylogenies constitute samples of the approximate posterior distribution.

Simulating Cophylogenies from the Prior Distribution

We simulate each cophylogeny for 2 treeducken-time, with treeducken-time being the arbitrary time unit defined in Dismukes and Heath (2021) that can represent any amount of

time in the real world (e.g., 1 Myr, 10 Myr, or 100 Myr). Thus, all simulated cophylogenies share the same time frame and differ only in speciation and extinction rates (Fig. 2). Despite sharing the same time frame, the simulations can be compared to a real cophylogeny of any time frame (in Myr) thanks to the arbitrary nature of the simulated time frame. Parameter values are drawn from a multivariate prior distribution:

$$\lambda_H \sim \text{Exp}(0.5), \lambda_S \sim \text{Exp}(0.5), \lambda_C \sim \text{Exp}(1.2), \lambda_W \sim \text{Exp}(0.5) \\ \epsilon_H, \epsilon_S \sim \text{Beta}(1,2)$$

The four speciation rates follow exponential distributions (Dismukes and Heath 2021), with theoretical means being 2, 2, 0.82, and 2 for λ_H , λ_S , λ_C , and λ_W , respectively. Because cospeciation involves speciation of both the hosts and symbionts and has a strong effect on cophylogeny size and computation time, the cospeciation rate is assigned a smaller theoretical mean to reduce the number of simulations that are timed out (each simulation is subject to a timeout of 600 s). The two relative extinction rates follow a Beta distribution with a decreasing probability density function ($\text{Beta}(1,2)$). The actual prior distribution is represented by 1,200,000 samples, which correspond to a set of 1,200,000 simulated cophylogenies. For simplicity and clarity, we refer to these simulated cophylogenies as the prior set of cophylogenies.

Distance Between Target and Simulated Cophylogenies

BLenD curves as summary statistics

Based on Branch Length Differences (BLenD), here we present two ways in which pairwise branch length differences can be sampled from a cophylogeny, which we will refer to as BLenD-a and BLenD-b. BLenD-a uses the branch length differences between associated host and symbiont tips as a measure of branch length congruence among tips, which is commonly used in cophylogenetic studies (Fig. 3a). For each host-symbiont association in the cophylogeny, we obtain:

$$\chi = \frac{l_h - l_s}{L}$$

where L is the crown of the host or symbiont phylogeny, whichever is longer (Fig. 2). Then, we can define a variable

$$x = (\chi_i - \chi_j)/2$$

Which is the pairwise difference between two samples of χ divided by 2 (to ensure x ranges between -1 and 1) for all possible pairs of χ . We perform kernel density estimation (KDE) on all samples of x to obtain $f(x)$, the density function of x :

$$f(x) = \text{KDE}(x_1, x_2, x_3, \dots)$$

In complement, BLenD-b uses the branch length differences between all unique pairs of host tips connected through a pair of sister symbionts (Fig. 3b). This extracts signal of coevolution, since the simultaneous branching of hosts and symbionts would create matching patterns of host-sisters and symbiont-sisters, giving shorter distances than random patterns of host-sisters and symbiont-sisters. For each pair of hosts, we obtain:

$$y = \frac{l_1 - l_2}{L}$$

where L is again the crown age of the host or symbiont phylogeny, whichever is longer (Fig. 2). We perform kernel density estimation (KDE) on all samples of y to obtain $g(y)$, the density function of y :

$$g(y) = KDE(y_1, y_2, y_3, \dots)$$

Importantly, if a single host tip is associated with a pair of sister symbionts, $l_1 = l_2$ and $y = 0$ (indicated by an asterisk * in Fig. 3b).

The kernel density estimation is performed with the function `density(kernel = "gaussian")` in the `stats` package in R (ver. 4.4.0; R Core Team (2024)). The bandwidth of such density estimation is handled automatically by the R function, but the bandwidth should be expected to decrease (and thus the accuracy of density estimation increases) as the cophylogeny gets larger. As density curves with the area under the curve equal to one, BLenD-a and BLenD-b curves are normalized and irrespective of the size of the cophylogeny.

BLenD-a and BLenD-b sample branch pairs from a cophylogeny in complementary ways, so we use them in combination to maximize the information that can be extracted from the cophylogeny of interest. Both BLenD-a and BLenD-b may sample multiple branch pairs from a "multiple association" (i.e., the association between one host and multiple symbionts or vice versa), allowing cophylogenies with multiple associations to be handled naturally (Fig. 3). Notably, the BLenD curves do not extract information from nodes deeper than the terminal branches, because the association between host and symbiont lineages becomes increasingly untraceable at deeper nodes.

The distance between the target cophylogeny and any simulated cophylogeny, based on the BLenD curves, is equal to the area between their respective BLenD curves (Fig. 4):

$$D_{BLenD-a}(simulated, target) = \int_{-1}^1 |f_{simulated}(x) - f_{target}(x)| dx$$

$$D_{BLenD-b}(simulated, target) = \int_{-1}^1 |g_{simulated}(y) - g_{target}(y)| dy$$

Similarly to Janzen et al. (2015), this definition of distance achieves a natural weighting of the contribution of the branch length differences even if the number of length differences is variable (i.e., regularization). In effect, the distance based on the BLenD curve is equal to zero only when the simulated BLenD curve is identical to the target BLenD curve.

256

257 Overall distance between target and simulated cophylogenies

258 To achieve higher informativeness, we combine the BLenD-a and BLenD-b curves with tree-size-
 259 based summary statistics, because speciation/extinction rates will impact the numbers of tips in
 260 the resultant cophylogeny. The summary statistics comprise four components:

- 261 1) BLenD-a curve
- 262 2) BLenD-b curve
- 263 3) Host tree size
- 264 4) Symbiont tree size

265 The tree size for the hosts is quantified as the log-transformed number of hosts in the extant
 266 host tree. Similarly, the tree size for the symbionts is the log-transformed number of symbionts
 267 in the extant symbiont tree. The distance between any simulated cophylogeny and the target
 268 cophylogeny based on host (or symbiont) tree size, $D_{host}(simulated, target)$ or
 269 $D_{symb}(simulated, target)$, is simply the absolute difference in host (or symbiont) tree size
 270 between the simulated and target cophylogenies. We standardize the summary statistics to
 271 ensure that each of the four components of summary statistics has the same standard
 272 deviation among all simulations. We then compute the overall distance between the simulated
 273 and target cophylogenies as

$$274 \quad D_{BLenD-a}(simulated, target) + D_{BLenD-b}(simulated, target) + D_{host}(simulated, target) \\
 275 \quad + D_{symb}(simulated, target)$$

276 In the ABC-based parameter estimation, simulated cophylogenies with the lowest overall
 277 distance to the target cophylogeny are accepted (the percentage of acceptance is termed the
 278 tolerance rate), and the parameters used to simulate the accepted cophylogenies constitute
 279 samples of the posterior distribution.

280 Importantly, none of the four components of summary statistics considers the time frame of
 281 the target cophylogeny. These summary statistics are essentially informative about the relative
 282 relationships (i.e., ratios) between the six speciation/extinction rates sharing the same absolute
 283 timescale, but they are not informative about the absolute timescale.

284

285 Two levels of inference: relative and absolute rates

286 In treeducken and thus in our method, the origination time of the cophylogeny is defined as
 287 the time point since which the two clades have been interacting with each other (Fig. 2). The
 288 time frame of the cophylogeny may be implied from the stem age of either interacting clade
 289 (e.g., the clade-to-clade symbiosis is as old as the stem group of symbionts), but such an
 290 implication may not be possible in some cases. Therefore, here we describe two levels of
 291 inference. A known time frame of the empirical cophylogeny is not necessary for the first level
 292 of inference and is only necessary for the second level of inference. This twofold approach

allows the method to be applied to time-calibrated cophylogenies whose time frame can or cannot be known.

The first level of inference estimates the speciation and extinction rates relative to the net diversification rate of the cophylogeny when the time frame of the cophylogeny is unknown (or not of interest). First, we define r , the net diversification rate of the cophylogeny, as the harmonic mean of r_{host} and $r_{symbiont}$

$$r = \frac{2 \cdot r_{host} \cdot r_{symbiont}}{r_{host} + r_{symbiont}} \quad (1)$$

where $r_{host} = \lambda_H + \lambda_C - \mu_H$ and $r_{symbiont} = \lambda_S + \lambda_C + \lambda_W - \mu_S$ are the net diversification rates of the host and symbiont clades, respectively. We then perform the first level of inference in the reparametrized space of $(\frac{\lambda_H}{r}, \frac{\lambda_S}{r}, \frac{\lambda_C}{r}, \frac{\lambda_W}{r}, \frac{\mu_H}{r}, \frac{\mu_S}{r})$ while being agnostic about $\lambda_H, \lambda_S, \lambda_C, \lambda_W, \mu_H, \mu_S$ and r . This first level of inference is timescale-agnostic: it does not require information on the time frame of the target cophylogeny and only requires the consistency of time units between the host and symbiont phylogenies; therefore, the inferable parameters $(\frac{\lambda_H}{r}, \frac{\lambda_S}{r}, \frac{\lambda_C}{r}, \frac{\lambda_W}{r}, \frac{\mu_H}{r}, \frac{\mu_S}{r})$ are relative rates (ratios) and do not contain timescale information.

The second level of inference estimates the absolute speciation and extinction rates when the time frame of the cophylogeny is known and of interest. Without using the time frame of the target cophylogeny, we can already estimate the “absolute” rates in units of events/lineage/treeducken-time, with treeducken-time being an arbitrary time unit that can represent any amount of time in the real world – this is achieved through the same procedure as the first level of inference, but in the parameter space of $(\lambda_H, \lambda_S, \lambda_C, \lambda_W, \mu_H, \mu_S)$. We perform unit conversion as a final step to obtain estimates of absolute speciation and extinction rates in units of events/lineage/Myr. The absolute rates are calculated as:

$$\hat{\lambda}_H, \hat{\lambda}_S, \hat{\lambda}_C, \hat{\lambda}_W, \hat{\mu}_H, \text{ or } \hat{\mu}_S \times \frac{\text{Simulated time frame (arbitrary)}}{\text{Target time frame (empirical, in Myr)}} \quad (2)$$

Unlike the first level of inference, this second level of inference is timescale-aware: it requires information on the time frame of the target cophylogeny in addition to the consistency of time units between the host and symbiont phylogenies; therefore, the inferable parameters $(\lambda_H, \lambda_S, \lambda_C, \lambda_W, \mu_H, \mu_S)$ are absolute rates and contain timescale information.

Performance Evaluation

We test the overall performance of the ABC-based parameter estimation with a tolerance rate of 1/40000 (also known as acceptance rate, see “ABC framework” in Methods) for a set of 1000 simulated cophylogenies, whose speciation and relative extinction rates $(\lambda_H, \lambda_S, \lambda_C, \lambda_W, \epsilon_H, \epsilon_S)$

are drawn from the prior distribution. This ensures that the test set reflects the prior beliefs about the speciation and extinction rates in a cophylogenetic system.

Accuracy and honesty about uncertainty

The relative error of the (prior or posterior) distribution is quantified as the mean absolute percent error

$$|\bar{\theta} - \theta|/\theta$$

where $\bar{\theta}$ is the mean of the (prior or posterior) distribution and θ is the true value. A high relative error indicates poor accuracy of the estimated mean. We compute the absolute z-score of the true value relative to the (prior or posterior) distribution as

$$|\theta - \bar{\theta}|/\hat{\sigma}$$

where $\hat{\sigma}$ is the standard deviation of the (prior or posterior) distribution. The absolute z-score is a measure of overconfidence, which is defined as the degree to which uncertainty (the spread of the distribution) is underestimated relative to actual error. A high z-score indicates strong overconfidence, whereas a low z-score indicates weak overconfidence and honesty about uncertainty.

Effects of cophylogeny size on performance

The size of the cophylogeny of interest, hereafter referred to as cophylogeny size for simplicity, is expected to affect performance. This is because larger cophylogenies should allow the BLend-a and BLend-b curves to be estimated from more samples (i.e., corresponding to smaller bandwidths in the kernel density estimation, see "BLend curves as summary statistics"). Following Hayward et al. (2021), cophylogeny size is quantified as the harmonic mean of the host tree size and symbiont tree size given by

$$\frac{2n_h n_s}{n_h + n_s}$$

where n_h and n_s are the numbers of tips for the host or symbiont tree, respectively (this is similar to the definition of the net diversification rate of the cophylogeny in Equation 1). The treeducken simulator does not allow us to simulate cophylogenies of a specific size (Dismukes and Heath 2021), but our test set of cophylogenies includes a wide variety of parameterizations (true speciation and extinction rates) drawn randomly from the prior distribution, allowing cophylogeny size to vary accordingly.

Application to Empirical Data

The ABC approach is applicable to time-calibrated cophylogenetic datasets, which allow the timeline to be held consistent for all six of the speciation/extinction processes. Here we

demonstrate the application of this method to empirical data with a time-calibrated cophylogeny of Batesian mimicry compiled by Van Dam et al. (2024), where beetles in the genus *Doliops* mimic beetles in the genus *Pachyrhynchus*, both native to the Philippines. This mimicry can be seen as a case of commensalism, which is defined as a type of interaction where one party in the interaction receives a benefit while the other party receives neither a cost nor a benefit (Bronstein 2015). Particularly in this case, the mimics (*Doliops*) receive a protective benefit while the models (*Pachyrhynchus*) are largely unaffected. Host-symbiont models have been fitted to this dataset in the original study (Van Dam et al. 2024), with mimics considered as symbionts and models as hosts. We use the `phylo.tracer` function from the `physketch` package (Revell 2020) to extract the two beetle phylogenies from Figure 4 of the original study while preserving their branch lengths in Myr. We then code the association between species manually based on Figure 2 of the original study.

This is done using the same ABC framework and the same prior set of simulated cophylogenies ($n=1,200,000$) that we use for performance evaluation and a tolerance rate (also known as acceptance rate, see “ABC Inferential Framework” in Methods) of $1/40000$. We also perform posterior predictive checks of model fit (Pantel and Becks 2023) to see how well the simulated cophylogenies that are accepted in the ABC framework (i.e., those that contribute to the approximate posterior distribution of the parameter estimates) resemble the *Pachyrhynchus-Doliops* mimicry cophylogeny.

Results

Performance

The posterior relative errors of both the first and second levels of inference tend to decrease with cophylogeny size (Fig. 5a). For the relative rates ($\frac{\lambda_H}{r}, \frac{\lambda_S}{r}, \frac{\lambda_C}{r}, \frac{\lambda_W}{r}, \frac{\mu_H}{r}, \frac{\mu_S}{r}$) in the first level of inference, the prior relative errors do not increase or decrease considerably with cophylogeny size, whereas the posterior relative errors decrease considerably with cophylogeny size. For the absolute rates ($\lambda_H, \lambda_S, \lambda_C, \lambda_W, \mu_H, \mu_S$) in the second level of inference, both the prior and posterior relative errors decrease considerably with cophylogeny size. When cophylogeny size is large, however, the relative rates and their respective absolute rates achieve almost equal posterior errors.

The posterior overconfidence of both levels of inference generally stays below 1 (75% percentile lines in Fig. 5b). This suggests that the standard deviation of the posterior distribution is generally equal to or greater than the actual absolute error and thus honestly reflects uncertainty. Notably, the posterior overconfidence of the relative rates ($\frac{\lambda_H}{r}, \frac{\lambda_S}{r}, \frac{\lambda_C}{r}, \frac{\lambda_W}{r}, \frac{\mu_H}{r}, \frac{\mu_S}{r}$) increases with cophylogeny size. When cophylogeny size is large, the relative rates and their respective absolute rates are almost equally overconfident (i.e., almost equally honest about uncertainty).

Error and overconfidence show consistent relationships with cophylogeny size. These patterns do not differ between speciation and extinction rates, although they correspond, respectively,

to birth and death processes in the co-diversification model. Overall, the ABC approach achieves clearly improved accuracy relative to the prior and reasonable honesty about uncertainty when the target cophylogeny is sufficiently large.

Here, we provide computation time estimates for the ABC parameter inference based on its performance on the Puma computing cluster at the University of Arizona. Time needed to simulate the prior set of 1,200,000 cophylogenies (see "Simulating Cophylogenies from the Prior Distribution" in Methods) depends on the assumed relative extinction rates (ϵ_H and ϵ_S), with the simulations taking longer to run when ϵ_H and ϵ_S are small. Simulations are run in parallel to ensure a sufficiently large sample size to cover the multivariate (6-dimensional) prior distribution. It takes at most 10 hours to simulate the prior set of cophylogenies when 5 jobs (each containing 91 parallel tasks running on 94 AMD EPYC 7642 CPU cores) are submitted in parallel. This is a time-consuming step, but the prior set of cophylogenies can be reused and does not need to be re-simulated every time the parameter inference is performed. The ABC procedure itself, not counting the simulation of the prior set of cophylogenies, takes less than 1 minute on average (i.e., our performance evaluation against a test set of 1000 cophylogenies is finished within 15 hours). However, the computation time for both the simulations and the ABC itself depends on the exact implementation and likely can be reduced through optimization of the cophylogeny simulator, data structure of the summary statistics, etc.

Example: *Pachyrhynchus-Doliops* Mimicry

The first level of inference (of the relative speciation and extinction rates), without using the time frame of the *Pachyrhynchus-Doliops* mimicry cophylogeny, gives the following estimates (Fig. 6b; with means and standard deviations reported here): $\frac{\lambda_H}{r}$ is estimated to be 1.90 ± 0.80 , $\frac{\lambda_S}{r}$ is estimated to be 0.87 ± 0.71 , $\frac{\lambda_C}{r}$ is estimated to be 0.33 ± 0.28 , $\frac{\lambda_W}{r}$ is estimated to be 0.62 ± 0.84 , $\frac{\mu_H}{r}$ is estimated to be 0.86 ± 0.82 , and $\frac{\mu_S}{r}$ is estimated to be 0.90 ± 1.06 . The joint distribution of the relative rates shows a nuanced picture of the correlation between the six absolute rates (Fig. S5).

For the second level of inference (of the absolute speciation and extinction rates), we use 27.5 Myr as the implied time frame of the *Pachyrhynchus-Doliops* mimicry cophylogeny in Equation 2. This assumes that the time frame of the model-mimic cophylogeny is as old as the stem age of the mimic clade *Doliops*, which is estimated to be 25–30 Myr (midpoint: 27.5 Myr) based on a geological calibration for the maximum age of the Philippine Islands to which *Doliops* is endemic (Van Dam et al. 2024). Given Equation (12) and an time frame of 2 treeducken-time for the simulated cophylogenies: λ_H is estimated to be 2.76 ± 0.99 events/lineage/Myr, λ_S is estimated to be 1.28 ± 0.96 events/lineage/Myr, λ_C is estimated to be 0.48 ± 0.35 events/lineage/Myr, λ_W is estimated to be 0.93 ± 1.24 events/lineage/Myr, μ_H is estimated to be 1.17 ± 1.10 events/lineage/Myr, and μ_S is estimated to be 1.24 ± 1.46 events/lineage/Myr. The joint distribution of the relative rates shows a nuanced picture of the correlation between the

six absolute rates (Fig. S6) similar but not identical to the correlation between the six relative rates (Fig. S5)

For both the first and second levels of inference, we cannot rule out the possibility that incomplete sampling (Van Dam et al. 2023; Van Dam et al. 2024) biases the speciation/extinction rate estimates. *treeduck*-simulated cophylogenies that are accepted in the ABC fit the *Pachyrhynchus-Doliops* mimicry cophylogeny reasonably well, as indicated by our posterior predictive checks for the BLenD-a curve, BLenD-b curve, the number of hosts, and the number of symbionts (Fig. 6d-6g).

Discussion

Here we present the proof of concept of an ABC approach for inferring, simultaneously, rates of different types of speciation and extinction from a cophylogenetic system. Key to our approach is the newly designed summary statistics based on the density curves of Branch Length Differences (BLenD). We show that the new ABC approach can handle several expected technical difficulties reasonably well. Specifically, (i) Stochasticity: both speciation and extinction are simulated as stochastic processes in the birth-death co-diversification model, introducing stochasticity to the resulting cophylogenies (Dismukes and Heath 2021). (ii) Incompleteness of information: an extant cophylogeny does not contain information about extinct lineages or the historical association between hosts and symbionts. Compared to previous approaches to estimating macroevolutionary rates of species interactions from cophylogenies (Alcala et al. 2017; Satler et al. 2019), our new ABC approach allows speciation/extinction rate estimation under a more biologically realistic assumption that speciation-extinction dynamics occur simultaneously in the host and symbiont clades on the same timeline.

Timescale: relative versus absolute rates

The first level of inference (timescale-agnostic) has relatively relaxed data requirements, whereas the second level of inference (timescale-aware) provides additional informativeness thanks to more stringent data requirements. Specifically, without information on the time frame of the cophylogeny, the first level of inference can estimate the six speciation and extinction rates relative to the net diversification rate, provided the host and symbiont phylogenies share the same time unit. In comparison, the second level of inference, using the time frame of the cophylogeny, adds another layer of information by estimating the absolute speciation/extinction rates in the realistic unit of events/lineage/Myr. When the objective is to quantify the contributions of different speciation/extinction processes to the net diversification *within* a single cophylogeny, the time frame of the cophylogeny can be considered not of interest as long as the same time frame is shared by all speciation/extinction processes; in this case, the first level of inference (of the relative rates) should suffice. When the objective is to compare speciation/extinction rates *across* different cophylogenetic systems, however, speciation/extinction rates in units of events/lineage/Myr are necessary, and so is the second level of inference (of the absolute rates). It is important to note that the time frame of the

cophylogeny may be uncertain and thus introduce uncertainty to the second level of inference, but not the first level of inference.

Indices versus normalized curves as summary statistics

The key to successful ABC parameter inference is summary statistics that decide the distance between a simulation and the observed data (Sunnåker et al. 2013). Since a cophylogenetic system consists of two phylogenies and the network formed by the tips of the two trees, one could, in theory, use indices of phylogenies and networks as the summary statistics for ABC. Indices for phylogenies include phylogenetic diversity (Clarke and Warwick 2001), the gamma statistic (Pybus and Harvey 2000) and Sackin index measuring tree shape (Sackin 1972). Indices for networks include global properties such as nestedness, modularity, connectance (Guimaraes Jr 2020), and meso-scale properties such as motif frequencies (Simmons et al. 2019). Indices concerning both the phylogeny and network components of a cophylogeny include the mantel correlation, a measure of phylogenetic conservatism of bipartite species association (Maliet et al. 2020). However, single indices tend not to be informative enough as summary statistics for phylogenetic systems (Janzen et al. 2015; Janzen and Etienne 2024). On the other hand, when we combine multiple of these indices in our preliminary exploratory analyses, finding reliable correlations between data patterns (summary statistics) and data-generating parameters becomes difficult. This difficulty in our preliminary analyses agrees with the previous study on estimating macroevolutionary rates of species interactions from cophylogenies, where accurate parameter inference based on numerous indices is difficult without dimension reduction and machine learning techniques (Alcala et al. 2017). Because of these reasons, we do not choose index-based summary statistics for our approach.

Indeed, the size-based summary statistics used in our approach are arguably index-based as well (in the sense that they are single numbers as opposed to normalized curves). To better focus on the newly designed BLenD curves as summary statistics, we use the simplest forms of size-based summary statistics (i.e., tree sizes) without a formal comparison to other choices of size-based summary statistics. However, we acknowledge that a single composite metric, such as the number of effective pairs $2n_h n_s / (n_h + n_s)$ (Hayward et al. 2021) or the number of extant associations in the host-symbiont network, might be equally promising choices for size-based summary statistics. These size-based summary statistics should not be ruled out by potential future efforts to improve upon the ABC shown here.

Normalized curves have been proposed as an alternative type of summary statistics for ABC. Specifically, a pioneering study has used a normalized lineage-through-time curve (nLTT), which is irrespective of the size or length of the phylogenetic tree, to effectively estimate constant speciation and extinction rates from single phylogenies while taking into account constant-rate lineage extinction (Janzen et al. 2015). The nLTT summary statistics are informative and computationally efficient. Inspired by this work, in this study we adapt the use of normalized curves as summary statistics for cophylogenetic systems.

A possible explanation for the effectiveness of the BLenD curves may be that they offer high informativeness with relatively low effective dimensionality. Density curves like the BLenD-a

and BLenD-b curves have an area-under-the-curve (AUC) of 1. This constraint, combined with the smoothness of kernel density curves, should mean that any two points on the same BLenD-a or BLenD-b curve are not fully independent of each other. Thus, the BLenD-a and BLenD-b curves should be able to be effectively represented in a finite-dimensional space. However, this remains merely an intuitively plausible explanation.

Importantly, we acknowledge that BLenD-a and BLenD-b are just two examples and that there may be other ways to sample branch length differences from a cophylogeny that fall under the broader definition of BLenD density curves (i.e., density curves of pairwise branch length differences).

Simplifications of the model and inference

1. Phylogenetic lability of host switches

Phylogenetic distance between hosts has been shown to be important in determining whether host switches are successful or not. For symbionts such as parasites, much evidence shows that host switches occur more often between closely-related species than between distantly-related ones (i.e., phylogenetic conservatism of host switches, (Charleston and Robertson 2002; Engelstädter and Hurst 2006; Longdon et al. 2014; Engelstädter and Fortuna 2019). Recent simulation studies have also incorporated the phylogenetic conservatism of host switches into their models (Braga et al. 2020, 2021). However, the treeducken simulator (Dismukes and Heath 2021) used in this study assumes that host switches are phylogenetically labile, in the sense that the probability of a host switch is independent of the phylogenetic distance between the old host and new host. Thus, incorporating the phylogenetic conservatism of host switches may improve the realism of the simulation model. However, doing so may come at the cost of parameter identifiability due to an increased model complexity.

2. Host switches through symbiont cladogenesis

The treeducken simulator used in this study offers two options to model changes in host use by symbionts, host switches and host expansions (Dismukes and Heath 2021). By treeducken's definition, a host switch is an event in which a new species (as a result of speciation) gains a new host while losing the old host, whereas a host expansion is an event in which the new species gains a new host while retaining the old host. Here we only consider the host switch option because it allows the breadth of hosts used by a symbiont lineage to remain constant instead of assuming that symbionts inevitably evolve to become increasingly generalized (i.e., using an increasingly wide range of hosts) over time, which is not always the case for symbionts (e.g., Hardy et al. 2020). However, we acknowledge that it is possible for a symbiont to expand (i.e., spread) to new hosts without losing its old host (e.g., Braga et al. 2020, even to phylogenetically distant hosts like in Drinkwater et al. 2016). An alternative approach would be to model host gains and losses separately, although this option is not available with the treeducken simulator used in this study.

Another simplification made in this study is that host switches happen necessarily as a result of the cladogenetic speciation (branching speciation) of symbionts. It is worth mentioning that host switches could also occur anagenetically in a symbiont lineage without involving speciation (e.g., Satler et al. 2019). Additional efforts would be needed to develop a way to infer the rates of these processes.

3. Constant speciation and extinction rates

A caveat of our method is that it assumes constant speciation and extinction rates over time and across the cophylogeny. The BLenD curves as summary statistics only extract information from the branch lengths of the tips of the host and symbiont phylogenies (see Methods for the details and rationale). Thus, in the presence of considerable heterogeneity in speciation and extinction rates over time, rate estimates from the ABC-based inference are better reflections of speciation and extinction rates at shallower nodes of the cophylogeny than deep ones. In theory, it should be challenging to develop inferential tools that allow speciation and extinction rates to vary over time, because the historical associations between hosts and symbionts (i.e., those at deeper nodes of the cophylogeny) are not known from extant cophylogenies. Thus, such endeavors may not lead to fruitful outcomes. Inference that allows speciation and extinction rates to vary among lineages, e.g., analogous to the inference of tip diversification rates for single clades (e.g., Vasconcelos et al. 2022), may be an interesting future direction. In such inference, however, it may be necessary to further simplify the birth-death model and reduce the number of parameters to avoid identifiability issues.

(Co)phylogenetic uncertainty

As with previous cophylogenetic studies, our ABC cophylogenetic approach is not immune to errors due to phylogenetic reconstruction or calibration, incomplete taxon sampling, or discord between species trees and gene trees (e.g., Ho and Phillips, 2009; Hughes et al. 2007; Onuferko et al. 2019). One solution to the problem of phylogenetic uncertainty is to perform cophylogenetic analyses on sets of trees from ML or Bayesian inferences (Pérez-Escobar et al. 2016; Balbuena et al. 2020; Llaberia-Robledillo et al. 2023). This solution is potentially applicable to our method thanks to its Bayesian framework; the posterior samples for all sets of trees could be pooled to form the ultimate posterior distribution. Additionally, apparent losses of symbionts in some host lineages may be the results of incomplete sampling of host-symbiont associations (i.e., cophylogenetic uncertainty, e.g., (Jackson and Charleston 2004; Charleston and Perkins 2006). This source of uncertainty may be more difficult to handle and may require the development of a new statistical framework for this specific purpose.

Cophylogenetics: reconciliation versus macroevolutionary rate estimation

An ever-growing and promising body of research has focused on reconciling the phylogeny of symbionts with that of their hosts (e.g., Merkle et al. 2010; Baudet et al. 2015; Sinaimeri et al. 2023; see Charleston and Perkins 2006; Charleston and Libeskind-Hadas 2014 for an overview). These studies map the symbiont phylogeny onto the host phylogeny to answer where

cospeciations, duplications, host switches, and symbiont losses occur along branches of the host phylogeny. As an intermediate step toward such reconciliation, recent approaches COALA and AmoCOALA have used ABC to estimate the rates of the aforementioned events (Baudet et al. 2015; Sinaimeri et al. 2023).

Here, we point out that these rates of events defined for reconciliation purposes in COALA and AmoCOALA are distinct from the speciation rates defined for clade diversification, which correspond to the rate parameter in a Poisson-like process in units of events per unit time (Ricklefs 2007) or events per lineage per unit time (Rabosky et al. 2007; Liow et al. 2010). In COALA and AmoCOALA, temporal constraints imposed by tree topologies (e.g., host switches are only possible between temporally coexisting species) are enforced only to the point where the *order* of events are feasible; this criterion is known as temporal feasibility (Stolzer et al. 2012). This means that COALA and AmoCOALA do not take into account branch lengths and are largely agnostic to the exact intervals between events, making it impossible to derive the rates of these events in units of events (per lineage) per unit time (i.e., akin to the problem of pseudocongruence in biogeographic inference, see Donoghue and Moore 2003).

Satler et al. (2019) went beyond reconciliation to quantify the relative contributions of host switching, pollinator speciation, and pollinator extinction to cophylogenetic pattern between figs and fig wasps. However, the DTL (Duplication, Transfer, Loss) model that Satler et al. (2019) adopted (Szöllősi et al. 2012, an extension of Tofigh 2009) was developed for reconciliation purposes and makes the simplifications that (1) time along the host tree is "coarse grained" such that duplication, transfer, and loss events are specified to occur in a given time slice corresponding to an interval of time between two successive speciation events and (2) all time slices are assumed to have equal width without considering the exact time intervals between speciation events. Such simplifications distort the exact timeline of events, preventing the approach of Satler et al. (2019) from estimating the rates of these events in units of events (per lineage) per unit time.

Therefore, while the rate estimates from the approaches of Baudet et al. (2015), Satler et al. (2019), and Sinaimeri et al. (2023) provide valuable insights into the probabilities of fundamental evolutionary and ecological events in a cophylogenetic system, these rates are incompatible with the commonly accepted definition of rates of events in the macroevolutionary literature. To the best of our knowledge, our study represents the first ABC-based cophylogenetic approach to offer a temporally explicit view of multiple types of macroevolutionary events.

Potential improvements

Several technical improvements would be potentially useful in order to go beyond the proof of concept that we show here. Single-layered feed-forward neural networks, a machine learning technique, have been used to improve the accuracy of ABC parameter inference (Csilléry et al. 2012). The improved accuracy is achieved through neural-network-based regression to correct rate estimates (Blum and François 2010). This approach currently only works with a few single summary statistics. Adapting this regression-based approach for the use of normalized curves

(such as BLenD) as summary statistics may be an interesting future direction. Another promising direction may be to enhance the ABC in this study with Sequential Monte Carlo (SMC), an algorithm that improves the accuracy of inference by iteratively generating new parameter samples from the posterior distribution and repeating the ABC procedure (for examples of ABC-SMC for (co)phylogenetic systems, see Janzen et al. 2015; Baudet et al. 2015; Sinimeri et al. 2023). It is even possible to go beyond an Approximate Bayesian Computation framework and use the highly informative BLenD curves as summary statistics for other simulation-based inference approaches, such as those based on machine learning. This may be a particularly promising future direction because machine learning models, once properly trained on sufficient data, are usually very fast to run (see Lambert et al. 2023, which includes a machine learning approach to macroevolutionary rate estimation using pre-selected summary statistics).

Broader implications

The potential user of this method might want to interpret the results from our approach in light of results from other methods. For instance, the rate estimates from our approach offer valuable information about conservatism, a popular topic in the biology of species interactions (Gómez et al. 2010). In symbiont speciation without host switching (λ_S), the descendant symbiont lineages retain associations with the host itself. Differently, in cospeciation (λ_C), the descendant symbiont lineages retain associations with the descendants of the ancestral host. In both symbiont speciation without host switching (λ_S) and cospeciation (λ_C), the descendant symbiont lineages retain their associations with the ancestral lineage (Fig. 1), contributing to host conservatism of the symbionts. On the contrary, in symbiont speciation with host switching (λ_W), one of the descendant symbiont lineages switches to a new host lineage, reducing the host conservatism of the symbionts (i.e., contributing to host lability). Thus, the simultaneous inference of λ_S , λ_C , and λ_W helps allow patterns of host conservatism to be attributed to two types of speciation increasing host conservatism and one type of speciation decreasing it. The approach presented here adds to the existing methods to help tease apart the relative contributions of different macroevolutionary processes to cophylogenetic patterns (Alcala et al. 2017; Satler et al. 2019), but under a more biologically realistic assumption that speciation and extinction occur on the same timeline in the host and symbiont clades. With this improved biological realism, different speciation processes' contributions may be more easily compared within the *same* cophylogenetic system; additionally, a certain speciation process's contributions may also be more easily compared between *multiple* cophylogenetic systems.

In the beetle mimicry example analyzed in this work, we find that the two types of speciation increasing host conservatism and the one type decreasing it do not differ in rate considerably (Fig. 4). Van Dam et al. (2024) reveals that, in the beetle mimicry system, a number of interactions are conserved over long periods of time while others are more labile and transient. This mixed pattern of host conservatism, as our results suggest, could have arisen from the fact that speciation events increasing host conservatism are balanced by those decreasing it.

When a cophylogeny is seen as an evolving system of its own, questions can be asked about what affects the rates of different types of speciation in a cophylogeny. In phylogenetic

comparative research, significant efforts have been devoted to studying how organismal traits affect speciation rates (Garamszegi 2014; Revell and Harmon 2022; Morlon et al. 2024). Similar questions can be asked about cophylogenetic systems. For example, key innovations have been hypothesized and debated as drivers of speciation for single clades (Miller et al. 2023) - do certain types of key innovation also affect the rates of speciation in a cophylogenetic system? In the beetle mimicry example, limited dispersal among islands has been shown to drive cospeciation (Van Dam et al. 2024). Thus, one hypothesis may be that the evolution of long-distance dispersal ability may decrease the rate of cospeciation. In a different vein, the old and still influential "escape-and-radiate" hypothesis states that the evolution of novel counter-defenses in symbionts allows them to "escape" from the host defenses and, as a result, radiate (Ehrlich and Raven 1964; Cogni et al. 2022). Then, what are the effects of these counter-defenses on symbiont speciation without host switching, cospeciation, and symbiont speciation with host switching? Comparing the speciation rates of cophylogenetic systems with these key innovations versus those without them may provide insights into these questions and hypotheses.

An interesting potential for the approach presented here lies in the possibility of informing real-world problems about public health. For instance, the coevolutionary history of coronaviruses and their mammalian hosts has been shown to be characterized by frequent host switches (Maestri et al. 2024). Comparisons of host-switching speciation rates between different strains of viruses (many of which are essentially clades) would allow one to ask whether highly virulent strains switch hosts more or less frequently than less virulent strains. Alternatively, one could ask whether historical rates of host-switching speciation predict pathogens' risk of spilling over to other host species in the future. These lines of research, however, would require knowledge on the origination times of the host-pathogen interactions of interest (and thus the time frames of the cophylogenies of interest).

Concluding remarks

By considering the cophylogenetic system as a whole, the ABC-based approach presented here estimates multiple types of speciation and extinction rates from a cophylogenetic system. Fortunately, time-calibrated cophylogenetic datasets that are needed for such inference, where both the host and symbiont trees are dated, are gradually gaining traction. For example, time-calibrated phylogenies of plants can be generated with a readily available tool V.PhyloMaker2 (Jin and Qian 2022). When combined with data on plant-animal networks as well as dated phylogenies of animals, V.PhyloMaker2 allows time-calibrated cophylogenies of plant-animal interactions to be compiled. Some cophylogenetic systems consisting of eukaryotes and non-eukaryotes have also been time-calibrated, such as one consisting of mammals and their coronaviruses (Maestri et al. 2024). As a remarkable recent advance in this direction, Wang and Luo (2025) successfully dated the Tree of Life for symbiotic bacteria, potentially allowing time-calibrated cophylogenies to be compiled for several important types of interactions between symbiotic bacteria (e.g., Rickettsiales, *Wolbachia*) and their eukaryotic hosts. Notably, an ambitious recent compilation has included time-calibrated cophylogenies for three different systems including plant-bat interactions (pollination), plant-bird interactions (seed dispersal), and mycorrhizal symbioses between plants and fungi (Perez-Lamarque et al. 2025).

Undoubtedly, the lack of knowledge about the origination time of many interactions (and thus about the time frame of the cophylogeny) still poses challenges to the inference of absolute speciation/extinction rates from cophylogenies. It is, however, as shown in the inference of relative speciation/extinction rates presented in this paper, possible to make meaningful inferences about speciation/extinction rates in cophylogenies to better understand questions in ecology and evolution.

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Supplementary Material

Supplementary figures and tables are attached to the main document and available at: <https://doi.org/10.5061/dryad.02v6wwqfr>.

Data Availability

Code used in this study is available at: https://github.com/yichaozeng/cophy_ABC. Datasets used in this study are available at: <https://doi.org/10.5061/dryad.02v6wwqfr>.

References

- Aguilée, Robin, Fanny Gascuel, Amaury Lambert, and Régis Fèrrière. 2018. "Clade Diversification Dynamics and the Biotic and Abiotic Controls of Speciation and Extinction Rates." *Nature Communications* 9 (1): 3013.
- Alcala, Nicolas, Tania Jenkins, Philippe Christe, and Séverine Vuilleumier. 2017. "Host Shift and Cospeciation Rate Estimation from Co-Phylogenies." *Ecology Letters* 20 (8): 1014–24.
- Balbuena, Juan Antonio, Óscar Alejandro Pérez-Escobar, Cristina Llopis-Belenguer, and Isabel Blasco-Costa. 2020. "Random Tanglegram Partitions (Random TaPas): An Alexandrian Approach to the Cophylogenetic Gordian Knot." *Systematic Biology* 69 (6): 1212–30.

- 744 Baudet, Christian, Béatrice Donati, Blerina Sinimeri, Pierluigi Crescenzi, Christian Gautier,
745 Catherine Matias, and M-F Sagot. 2015. "Cophylogeny Reconstruction via an Approximate
746 Bayesian Computation." *Systematic Biology* 64 (3): 416–31.
- 747 Blasco-Costa, Isabel, Alexander Hayward, Robert Poulin, and Juan A Balbuena. 2021. "Next-
748 Generation Cophylogeny: Unravelling Eco-Evolutionary Processes." *Trends in Ecology &
749 Evolution* 36 (10): 907–18.
- 750 Blum, Michael GB, and Olivier François. 2010. "Non-Linear Regression Models for Approximate
751 Bayesian Computation." *Statistics and Computing* 20: 63–73.
- 752 Braga, Mariana P, Niklas Janz, Sören Nylin, Fredrik Ronquist, and Michael J Landis. 2021.
753 "Phylogenetic Reconstruction of Ancestral Ecological Networks Through Time for Pierid
754 Butterflies and Their Host Plants." *Ecology Letters* 24 (10): 2134–45.
- 755 Braga, Mariana P, Michael J Landis, Sören Nylin, Niklas Janz, and Fredrik Ronquist. 2020.
756 "Bayesian Inference of Ancestral Host–Parasite Interactions Under a Phylogenetic Model of
757 Host Repertoire Evolution." *Systematic Biology* 69 (6): 1149–62.
- 758 Bronstein, Judith L. 2015. *Mutualism*. Oxford University Press.
- 759 Burin, Gustavo, Paulo R Guimaraes Jr, and Tiago B Quental. 2021. "Macroevolutionary Stability
760 Predicts Interaction Patterns of Species in Seed Dispersal Networks." *Science* 372 (6543): 733–
761 37.
- 762 Chaparro-Pedraza, P Catalina, Gregory Roth, and Ole Seehausen. 2022. "The Enrichment
763 Paradox in Adaptive Radiations: Emergence of Predators Hinders Diversification in Resource
764 Rich Environments." *Ecology Letters* 25 (4): 802–13.
- 765 Charleston, Michael A, and Susan L Perkins. 2006. "Traversing the Tangle: Algorithms and
766 Applications for Cophylogenetic Studies." *Journal of Biomedical Informatics* 39 (1): 62–71.
- 767 Charleston, Michael A, and David L Robertson. 2002. "Preferential Host Switching by Primate
768 Lentiviruses Can Account for Phylogenetic Similarity with the Primate Phylogeny." *Systematic
769 Biology* 51 (3): 528–35.
- 770 Charleston, Michael, and Ran Libeskind-Hadas. 2014. "Event-Based Cophylogenetic
771 Comparative Analysis." In *Modern Phylogenetic Comparative Methods and Their Application in
772 Evolutionary Biology: Concepts and Practice*, edited by László Zsolt Garamszegi, 465–80.
773 Springer.
- 774 Clarke, KR, and RM Warwick. 2001. "A Further Biodiversity Index Applicable to Species Lists:
775 Variation in Taxonomic Distinctness." *Marine Ecology Progress Series* 216: 265–78.
- 776 Coelho, Marco Túlio P, and Thiago F Rangel. 2018. "Neutral Community Dynamics and the
777 Evolution of Species Interactions." *The American Naturalist* 191 (4): 421–34.

- 778 Cogni, Rodrigo, Tiago B Quental, and Paulo R Guimarães Jr. 2022. "Ehrlich and Raven Escape
779 and Radiate Coevolution Hypothesis at Different Levels of Organization: Past and Future
780 Perspectives." *Evolution* 76 (6): 1108–23.
- 781 Csilléry, Katalin, Michael GB Blum, Oscar E Gaggiotti, and Olivier François. 2010. "Approximate
782 Bayesian Computation (ABC) in Practice." *Trends in Ecology & Evolution* 25 (7): 410–18.
- 783 Csilléry, Katalin, Olivier François, and Michael GB Blum. 2012. "Abc: An r Package for
784 Approximate Bayesian Computation (ABC)." *Methods in Ecology and Evolution* 3 (3): 475–79.
- 785 Darwin, Charles. 1859. *On the Origin of Species by Means of Natural Selection*. London: John
786 Murray.
- 787 Dismukes, Wade, Mariana P Braga, David H Hembry, Tracy A Heath, and Michael J Landis. 2022.
788 "Cophylogenetic Methods to Untangle the Evolutionary History of Ecological Interactions."
789 *Annual Review of Ecology, Evolution, and Systematics* 53 (1): 275–98.
- 790 Dismukes, Wade, and Tracy A Heath. 2021. "Treeducken: An r Package for Simulating
791 Cophylogenetic Systems." *Methods in Ecology and Evolution* 12 (8): 1358–64.
- 792 Drinkwater, Benjamin, Angela Qiao, and Michael A Charleston. 2016. "WiSPA: A New Approach
793 for Dealing with Widespread Parasitism." *arXiv Preprint arXiv:1603.09415*.
- 794 Donoghue, Michael J, and Brian R Moore. 2003. "Toward an integrative historical
795 biogeography." *Integrative and Comparative Biology* 43 (2), 261–270.
- 796 Ehrlich, Paul R, and Peter H Raven. 1964. "Butterflies and plants: a study in coevolution."
797 *Evolution* 18: 586–608.
- 798 Engelstädter, Jan, and Nicole Z Fortuna. 2019. "The Dynamics of Preferential Host Switching:
799 Host Phylogeny as a Key Predictor of Parasite Distribution." *Evolution* 73 (7): 1330–40.
- 800 Engelstädter, Jan, and Gregory DD Hurst. 2006. "The Dynamics of Parasite Incidence Across
801 Host Species." *Evolutionary Ecology* 20 (6): 603–16.
- 802 Garamszegi, László Zsolt. 2014. *Modern Phylogenetic Comparative Methods and Their
803 Application in Evolutionary Biology: Concepts and Practice*. Springer.
- 804 Gómez, José M, Miguel Verdú, and Francisco Perfectti. 2010. "Ecological Interactions Are
805 Evolutionarily Conserved Across the Entire Tree of Life." *Nature* 465 (7300): 918–21.
- 806 Guimaraes Jr, Paulo R. 2020. "The Structure of Ecological Networks Across Levels of
807 Organization." *Annual Review of Ecology, Evolution, and Systematics* 51 (1): 433–60.
- 808 Hardy, Naté B, Chloe Kaczvinsky, Gwendolyn Bird, and Benjamin B Normark. 2020. "What We
809 Don't Know about Diet-Breadth Evolution in Herbivorous Insects." *Annual Review of Ecology,
810 Evolution, and Systematics* 51 (1): 103–22.

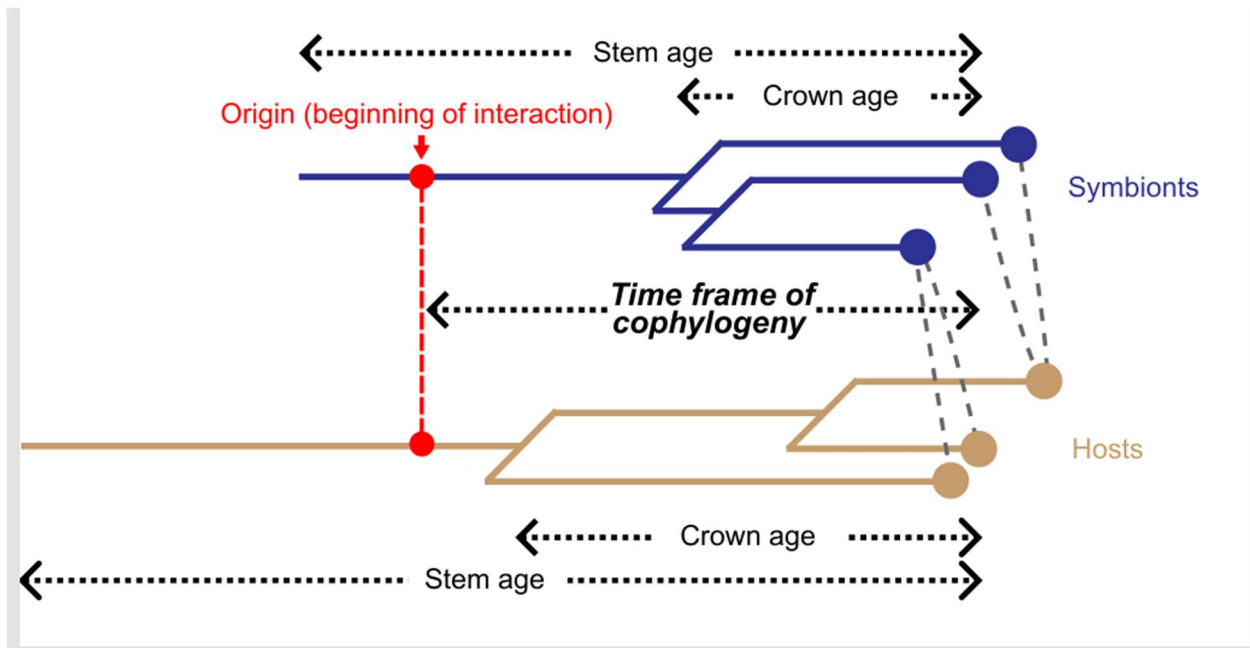
- 811 Harmon, Luke J, Cecilia S Andreazzi, Florence Débarre, Jonathan Drury, Emma E Goldberg,
812 Ayana B Martins, Carlos J Melián, et al. 2019. "Detecting the Macroevolutionary Signal of
813 Species Interactions." *Journal of Evolutionary Biology* 32 (8): 769–82.
- 814 Harmon, Luke J, JB Losos, DA Baum, D Futuyma, HE Hoekstra, RE Lenski, AJ Moore, CL Peichel, D
815 Schluter, and MC Whitlock. 2014. "Macroevolutionary Rates." *The Princeton Guide of Evolution*,
816 567–72.
- 817 Hayward, Alexander, Robert Poulin, and Shinichi Nakagawa. 2021. "A BROADSCALE ANALYSIS OF
818 HOST-SYMBIONT COPHYLOGENY REVEALS THE DRIVERS OF PHYLOGENETIC CONGRUENCE." *Ecology Letters*
819 24 (8): 1681–96.
- 820 Hembry, David H, and Marjorie G Weber. 2020. "Ecological Interactions and Macroevolution: A
821 New Field with Old Roots." *Annual Review of Ecology, Evolution, and Systematics* 51 (1): 215–
822 43.
- 823 Hembry, David H, Jeremy B Yoder, and Kari Roesch Goodman. 2014. "Coevolution and the
824 Diversification of Life." *The American Naturalist* 184 (4): 425–38.
- 825 Ho, S. Y. and M. J. Phillips. 2009. Accounting for calibration uncertainty in phylogenetic
826 estimation of evolutionary divergence times. *Systematic biology* 58:367–380. 94
- 827 Hoyal Cuthill, Jennifer, and Michael Charleston. 2012. "Phylogenetic Codivergence Supports
828 Coevolution of Mimetic Heliconius Butterflies." *PloS One* 7 (5): e36464.
- 829 Hughes, Joseph, Martyn Kennedy, Kevin P Johnson, Ricardo L Palma, and Roderic DM Page.
830 2007. "Multiple Cophylogenetic Analyses Reveal Frequent Cospeciation Between Pelecaniform
831 Birds and Pectinopygus Lice." *Systematic Biology* 56 (2): 232–51.
- 832 Jablonski, David. 2008. "Biotic Interactions and Macroevolution: Extensions and Mismatches
833 Across Scales and Levels." *Evolution* 62 (4): 715–39.
- 834 Jackson, Andrew P, and Michael A Charleston. 2004. "A Cophylogenetic Perspective of RNA–
835 Virus Evolution." *Molecular Biology and Evolution* 21 (1): 45–57.
- 836 Janzen, Thijs, and Rampal S Etienne. 2024. "Phylogenetic Tree Statistics: A Systematic Overview
837 Using the New r Package 'Treestats'." *Molecular Phylogenetics and Evolution*, 108168.
- 838 Janzen, Thijs, Sebastian Höhna, and Rampal S Etienne. 2015. "Approximate Bayesian
839 Computation of Diversification Rates from Molecular Phylogenies: Introducing a New Efficient
840 Summary Statistic, the nLTT." *Methods in Ecology and Evolution* 6 (5): 566–75.
- 841 Jin, Yi, and Hong Qian. 2022. "V. PhyloMaker2: An Updated and Enlarged r Package That Can
842 Generate Very Large Phylogenies for Vascular Plants." *Plant Diversity* 44 (4): 335–39.
- 843 Kaur, Katrina M, and Matt Pennell. 2023. "Synthesizing the Phylogenetic Evidence for
844 Mutualism-Associated Diversification." *Evolution* 77 (8): 1882–92.

- 845 Lambert, Sophia, Jakub Voznica, and H       Morlon. 2023. "Deep Learning from Phylogenies
846 for Diversification Analyses." *Systematic Biology* 72 (6): 1262  79.
- 847 Legendre, Pierre, Yves Desdevises, and Eric Bazin. 2002. "A Statistical Test for Host  Parasite
848 Coevolution." *Systematic Biology* 51 (2): 217  34.
- 849 Liow, Lee Hsiang, Tiago B Quental, and Charles R Marshall. 2010. "When Can Decreasing
850 Diversification Rates Be Detected with Molecular Phylogenies and the Fossil Record?"
851 *Systematic Biology* 59 (6): 646  59.
- 852 Llaberia-Robledillo, Mar, J Ignacio Lucas-Lled  , Oscar A P          , Boris R Krasnov, and
853 Juan Antonio Balbuena. 2023. "Rtapas: An r Package to Assess Cophylogenetic Signal Between
854 Two Evolutionary Histories." *Systematic Biology* 72 (4): 946  54.
- 855 Longdon, Ben, Michael A Brockhurst, Colin A Russell, John J Welch, and Francis M Jiggins. 2014.
856 "The Evolution and Genetics of Virus Host Shifts." *PLoS Pathogens* 10 (11): e1004395.
- 857 Maestri, Renan, Beno              , Anna Zhukova, and H       Morlon. 2024. "Recent
858 Evolutionary Origin and Localized Diversity Hotspots of Mammalian Coronaviruses." *Elife* 13:
859 RP91745.
- 860 Magallon, Susana, and Michael J Sanderson. 2001. "Absolute Diversification Rates in
861 Angiosperm Clades." *Evolution* 55 (9): 1762  80.
- 862 Maliet, Odile, Nicolas Loeuille, and H       Morlon. 2020. "An Individual-Based Model for the
863 Eco-Evolutionary Emergence of Bipartite Interaction Networks." *Ecology Letters* 23 (11): 1623  
864 34.
- 865 Marin, Jean-Michel, Pierre Pudlo, Christian P Robert, and Robin J Ryder. 2012. "Approximate
866 Bayesian Computational Methods." *Statistics and Computing* 22 (6): 1167  80.
- 867 Mart          , Andr    . 2016. "Phylogenetic Framework for Coevolutionary Studies: A
868 Compass for Exploring Jungles of Tangled Trees." *Current Zoology* 62 (4): 393  403.
- 869 Menet, Hugo, Vincent Daubin, and Eric Tannier. 2022. "Phylogenetic Reconciliation." *PLOS*
870 *Computational Biology* 18 (11): e1010621.
- 871 Merkle, Daniel, Martin Middendorf, and Nicolas Wieseke. 2010. "A Parameter-Adaptive
872 Dynamic Programming Approach for Inferring Cophylogenies." *BMC Bioinformatics* 11: 1  10.
- 873 Miller, Aryeh H, James T Stroud, and Jonathan B Losos. 2023. "The Ecology and Evolution of Key
874 Innovations." *Trends in Ecology & Evolution* 38 (2): 122  31.
- 875 Morlon, H      , J       Andr        , Jo     Barido-Sottani, Sophia Lambert, Beno            ,
876 Ignacio Quintero, Viktor Senderov, and Pierre Veron. 2024. "Phylogenetic Insights
877 into Diversification." *Annual Review of Ecology, Evolution, and Systematics* 55.

- 878 Morlon, H      , St       Robin, and Florian Hartig. 2022. "Studying Speciation and Extinction
879 Dynamics from Phylogenies: Addressing Identifiability Issues." *Trends in Ecology & Evolution* 37
880 (6): 497–506.
- 881 Nieberding, Caroline M, and Isabelle Olivieri. 2007. "Parasites: Proxies for Host Genealogy and
882 Ecology?" *Trends in Ecology & Evolution* 22 (3): 156–65.
- 883 Onuferko, Thomas M, Petr Bogusch, Rafael R Ferrari, and Laurence Packer. 2019. "Phylogeny
884 and Biogeography of the Cleptoparasitic Bee Genus *Epeolus* (Hymenoptera: Apidae) and
885 Cophylogenetic Analysis with Its Host Bee Genus *Colletes* (Hymenoptera: Colletidae)." *Molecular Phylogenetics and Evolution* 141: 106603.
- 887 Pantel, Jelena H, and Lutz Becks. 2023. "Statistical Methods to Identify Mechanisms in Studies
888 of Eco-Evolutionary Dynamics." *Trends in Ecology & Evolution* 38 (8): 760–72.
- 889 P          , Oscar Alejandro, Juan Antonio Balbuena, and Marc Gottschling. 2016. "Rumbling
890 Orchids: How to Assess Divergent Evolution Between Chloroplast Endosymbionts and the
891 Nuclear Host." *Systematic Biology* 65 (1): 51–65.
- 892 Perez-Lamarque, Benoit, Jeremy Andreoletti, Baptiste Morillon, Orane Pion-Piola, Amaury
893 Lambert, and Helene Morlon. 2025. "Darwin's Entangled Bank Through Deep Time: Structural
894 Stability of Mutualistic Networks over Evolutionary Timescales." *bioRxiv*, 2025–10.
- 895 Perez-Lamarque, Beno    , and H       Morlon. 2024. "Distinguishing Cophylogenetic Signal from
896 Phylogenetic Congruence Clarifies the Interplay Between Evolutionary History and Species
897 Interactions." *Systematic Biology* 73 (3): 613–22.
- 898 Pontarp, Mikael,      Br          , and Owen L Petchey. 2019. "Inferring Community Assembly
899 Processes from Macroscopic Patterns Using Dynamic Eco-Evolutionary Models and
900 Approximate Bayesian Computation (ABC)." *Methods in Ecology and Evolution* 10 (4): 450–60.
- 901 Pontarp, Mikael, Per Lundberg, and J       Ripa. 2024. "The Succession of Ecological Divergence
902 and Reproductive Isolation in Adaptive Radiations." *Journal of Theoretical Biology* 587: 111819.
- 903 Pybus, Oliver G, and Paul H Harvey. 2000. "Testing Macro-Evolutionary Models Using
904 Incomplete Molecular Phylogenies." *Proceedings of the Royal Society of London. Series B:
905 Biological Sciences* 267 (1459): 2267–72.
- 906 R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria:
907 R Foundation for Statistical Computing. <https://www.R-project.org/>.
- 908 Rabosky, Daniel L, Stephen C Donnellan, Amanda L Talaba, and Irby J Lovette. 2007.
909 "Exceptional Among-Lineage Variation in Diversification Rates During the Radiation of
910 Australia's Most Diverse Vertebrate Clade." *Proceedings of the Royal Society B: Biological
911 Sciences* 274 (1628): 2915–23.
- 912 Revell, Liam J. 2020. *PhyloSketch: Drawing Phylogenetic Objects Free-Hand*.
913 <http://github.com/liamrevell/phyloSketch>.

- 914 Revell, Liam J, and Luke J Harmon. 2022. *Phylogenetic Comparative Methods in R*. Princeton
915 University Press.
- 916 Ricklefs, Robert E. 2007. "Estimating Diversification Rates from Phylogenetic Information."
917 *Trends in Ecology & Evolution* 22 (11): 601–10.
- 918 Sackin, Michael J. 1972. "'Good' and 'Bad' Phenograms." *Systematic Biology* 21 (2): 225–26.
- 919 Satler, Jordan D, Edward Allen Herre, K Charlotte Jandér, Deren AR Eaton, Carlos A Machado,
920 Tracy A Heath, and John D Nason. 2019. "Inferring Processes of Coevolutionary Diversification
921 in a Community of Panamanian Strangler Figs and Associated Pollinating Wasps." *Evolution* 73
922 (11): 2295–2311.
- 923 Simmons, Benno I, Michelle JM Sweering, Maybritt Schillinger, Lynn V Dicks, William J
924 Sutherland, and Riccardo Di Clemente. 2019. "Bmotif: A Package for Motif Analyses of Bipartite
925 Networks." *Methods in Ecology and Evolution* 10 (5): 695–701.
- 926 Sinimeri, Blerina, Laura Urbini, Marie-France Sagot, and Catherine Matias. 2023. "Cophylogeny
927 Reconstruction Allowing for Multiple Associations Through Approximate Bayesian
928 Computation." *Systematic Biology* 72 (6): 1370–86.
- 929 Stadler, Tanja. 2013. "How Can We Improve Accuracy of Macroevolutionary Rate Estimates?"
930 *Systematic Biology* 62 (2): 321–29.
- 931 Stolzer, Maureen, Han Lai, Minli Xu, Deepa Sathaye, Benjamin Vernot, and Dannie Durand.
932 2012. "Inferring Duplications, Losses, Transfers and Incomplete Lineage Sorting with Nonbinary
933 Species Trees." *Bioinformatics* 28 (18): i409–15.
- 934 Sunnåker, Mikael, Alberto Giovanni Busetto, Elina Numminen, Jukka Corander, Matthieu Foll,
935 and Christophe Dessimoz. 2013. "Approximate Bayesian Computation." *PLoS Computational
936 Biology* 9 (1): e1002803.
- 937 Szöllősi, Gergely J, Bastien Boussau, Sophie S Abby, Eric Tannier, and Vincent Daubin. 2012.
938 "Phylogenetic Modeling of Lateral Gene Transfer Reconstructs the Pattern and Relative Timing
939 of Speciations." *Proceedings of the National Academy of Sciences* 109 (43): 17513–18.
- 940 Tofigh, Ali. 2009. "Using Trees to Capture Reticulate Evolution: Lateral Gene Transfers and
941 Cancer Progression." PhD thesis, KTH.
- 942 Van Dam, Matthew H., Analyn Anzano Cabras, and Athena W. Lam. 2023. "How the easter egg
943 weevils got their spots: Phylogenomics reveals müllerian mimicry in Pachyrhynchus
944 (Coleoptera, Curculionidae)." *Systematic Biology* 72(3): 516-529.
- 945 Van Dam, Matthew H, Alessandro Parisotto, Milton N Medina, Analyn A Cabras, Nayeli
946 Gutiérrez-Trejo, Bodo D Wilts, and Athena W Lam. 2024. "Biogeography Confounds the Signal
947 of Cospeciation in Batesian Mimicry." *Current Biology* 34 (23): 5554–63.

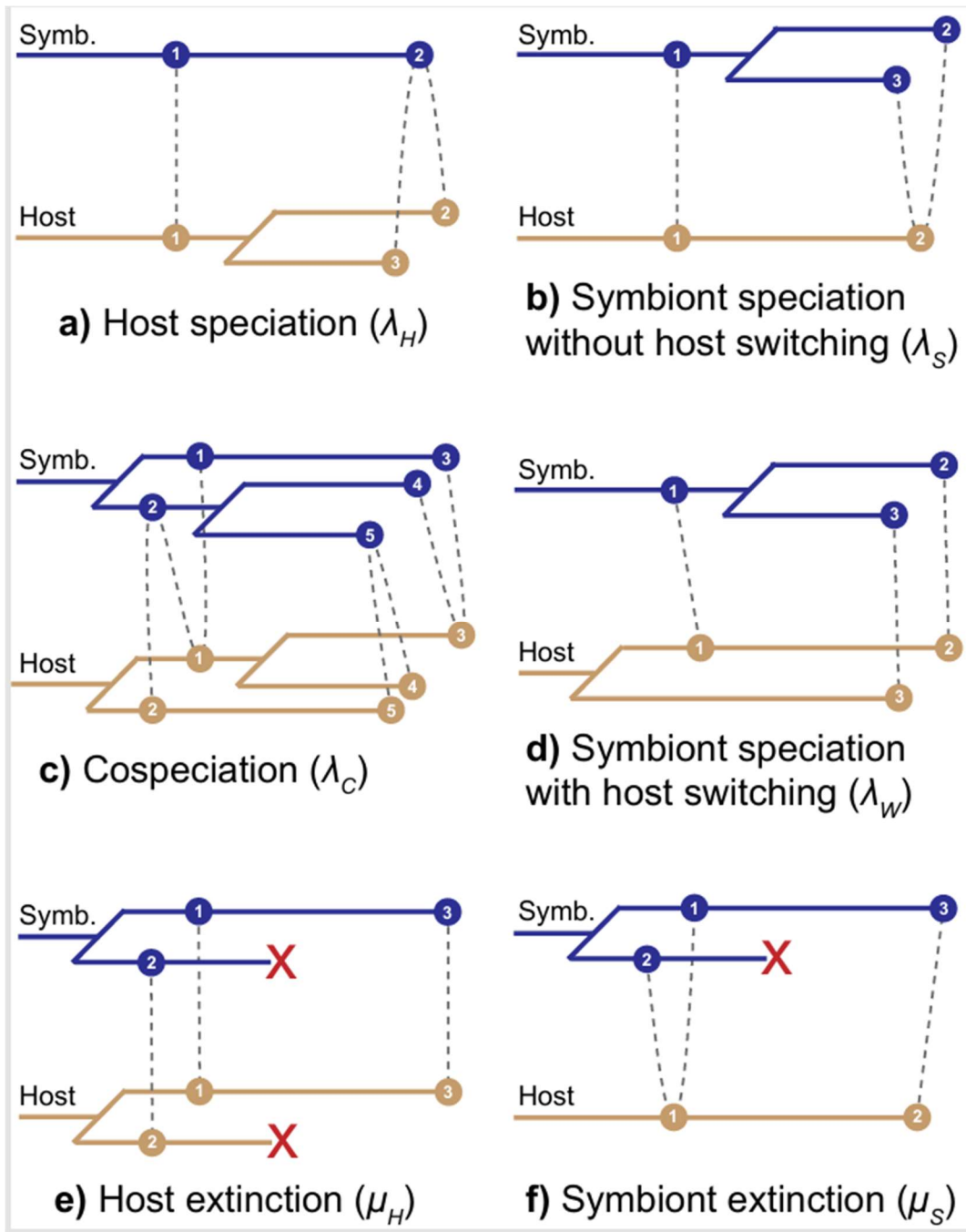
- 948 Vasconcelos, Thais, Brian C O'Meara, and Jeremy M Beaulieu. 2022. "A Flexible Method for
949 Estimating Tip Diversification Rates Across a Range of Speciation and Extinction Scenarios."
950 *Evolution* 76 (7): 1420–33.
- 951 Vienne, Damien M de, Tatiana Giraud, and Olivier C Martin. 2007. "A Congruence Index for
952 Testing Topological Similarity Between Trees." *Bioinformatics* 23 (23): 3119–24.
- 953 Vienne, DM de, G Refrégier, M López-Villavicencio, A Tellier, ME Hood, and TJNP Giraud. 2013.
954 "Cospeciation Vs Host-Shift Speciation: Methods for Testing, Evidence from Natural
955 Associations and Relation to Coevolution." *New Phytologist* 198 (2): 347–85.
- 956 Wang, Sishuo, and Haiwei Luo. 2025. "Dating the Bacterial Tree of Life Based on Ancient
957 Symbiosis." *Systematic Biology*, syae071.
- 958 Yoder, Jeremy B, and Scott L Nuismer. 2010. "When Does Coevolution Promote
959 Diversification?" *The American Naturalist* 176 (6): 802–17.
- 960 Zeng, Yichao, and David H Hembry. 2024. "Coevolution-Induced Selection for and Against
961 Phenotypic Novelty Shapes Species Richness in Clade Co-Diversification." *Journal of*
962 *Evolutionary Biology* 37 (12): 1510–22.
- 963 Zeng, Yichao, and John J Wiens. 2021. "Species Interactions Have Predictable Impacts on
964 Diversification." *Ecology Letters* 24 (2): 239–48.
- 965



967

968 *Figure 1. Important lengths (representations of time) in a cophylogeny. The origin of the*
 969 *cophylogeny is the time point since which the two clades have been interacting with each other.*
 970 *treeducken, the birth-death model used in this study, simulates the joint diversification history*
 971 *of the two interacting clades since the beginning of the interaction, but it does not preclude*
 972 *situations in which the two clades existed before the interaction originated. In reality, a time-*
 973 *calibrated cophylogeny may not contain information about the time frame of the cophylogeny,*
 974 *but it inherently contains information about the crown age of each interacting clade, which is*
 975 *used by the method described in this study for the inference of speciation and extinction rates.*

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978

979 *Figure 2. Speciation and extinction events giving rise to cophylogenetic patterns and their rates*
 980 *(in parentheses). (a) Host speciation (λ_H): both descendant hosts retain association with the*
 981 *original symbiont. (b) Symbiont speciation without host switching (λ_S): both descendant*
 982 *symbionts retain association with the original host. (c) Cospeciation (λ_C): a host and one of its*
 983 *symbionts, randomly selected, undergo speciation simultaneously (host 1 into hosts 3 and 4;*
 984 *symbiont 2 into symbionts 4 and 5). Each descendant host forms association with one of the*
 985 *descendant symbionts (host 3 with symbiont 4; host 4 with symbiont 5). The ancestral host's*

and symbiont's remaining associations are randomly sorted among the descendants. **(d)** Symbiont speciation with host switching (λ_W): same as (b), but here one of the descendant symbionts switches to a new, randomly selected host. **(e)** Host extinction (μ_H): note that the symbiont(s) relying on this host go extinct as well. **(f)** Symbiont extinction (μ_S). These event definitions are identical to those in the simulator, except that the rate of symbiont speciation with host switching (λ_W) is denoted by χ in the original study (Dismukes and Heath 2021). The rate of each type of event is defined as the number of events per lineage per unit time. Specifically, the rate of cospeciation (λ_C) is defined as the number of events per unit time per host lineage (i.e., in each cospeciation event, the speciating symbiont is selected at random among all symbionts associated with the host lineage).

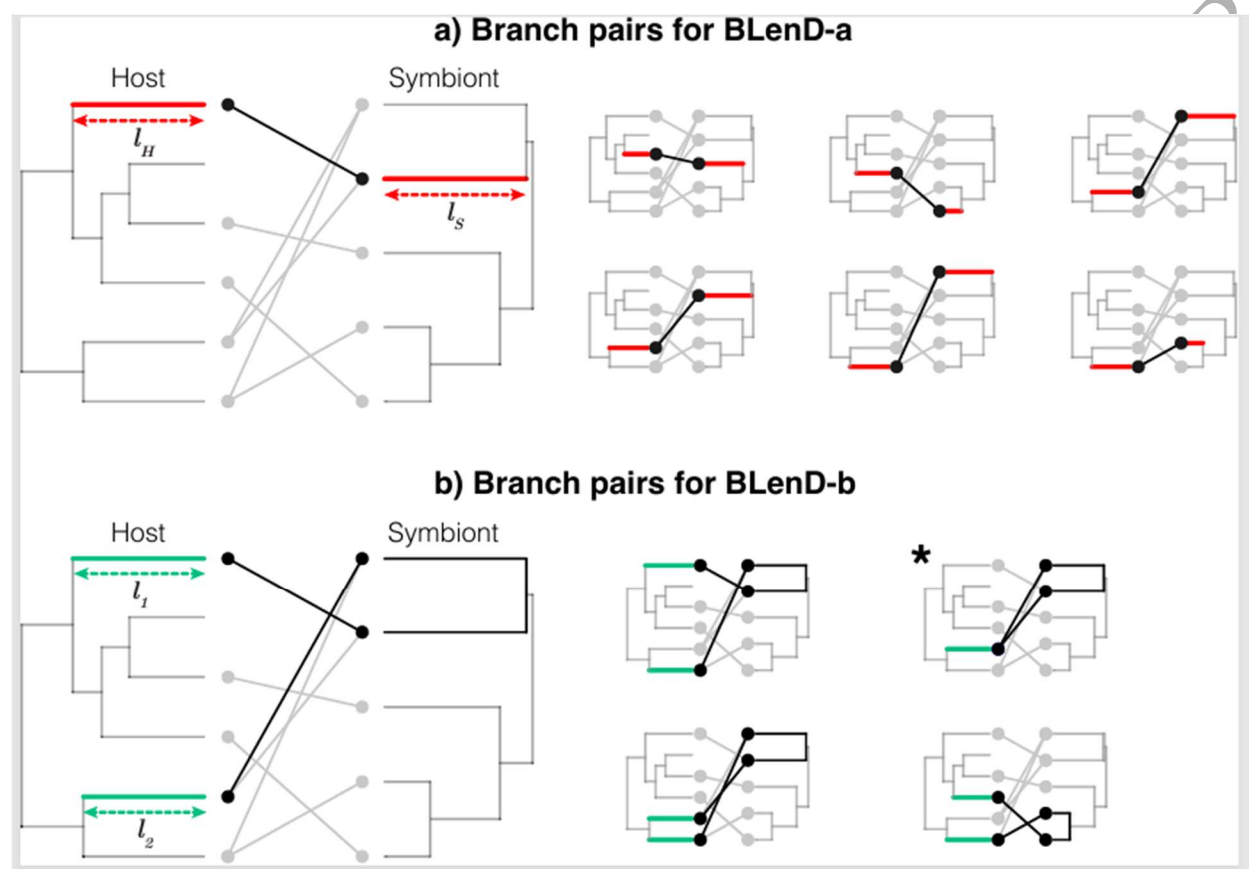


Figure 3. The pairing between branches for the computation of Branch Length Differences (BLenD). BLenD-a and BLenD-b refer to two ways in which branches are paired. **(a)**: in BLenD-a, terminal branches are paired for each symbiont-host association (l_h & l_s). **(b)**: in BLenD-b, terminal branches are paired for each unique pair of host tips connected through a pair of sister symbionts (l_1 & l_2). Importantly, if a single host tip is associated with a pair of sister symbionts, BLenD-b considers that host to be paired with itself and $l_1 = l_2$, as indicated by an asterisk in (b). Notably, neither BLenD-a nor BLenD-b considers information from nodes deeper than the terminal branches, because the association between host and symbiont lineages becomes increasingly untraceable at deeper nodes.

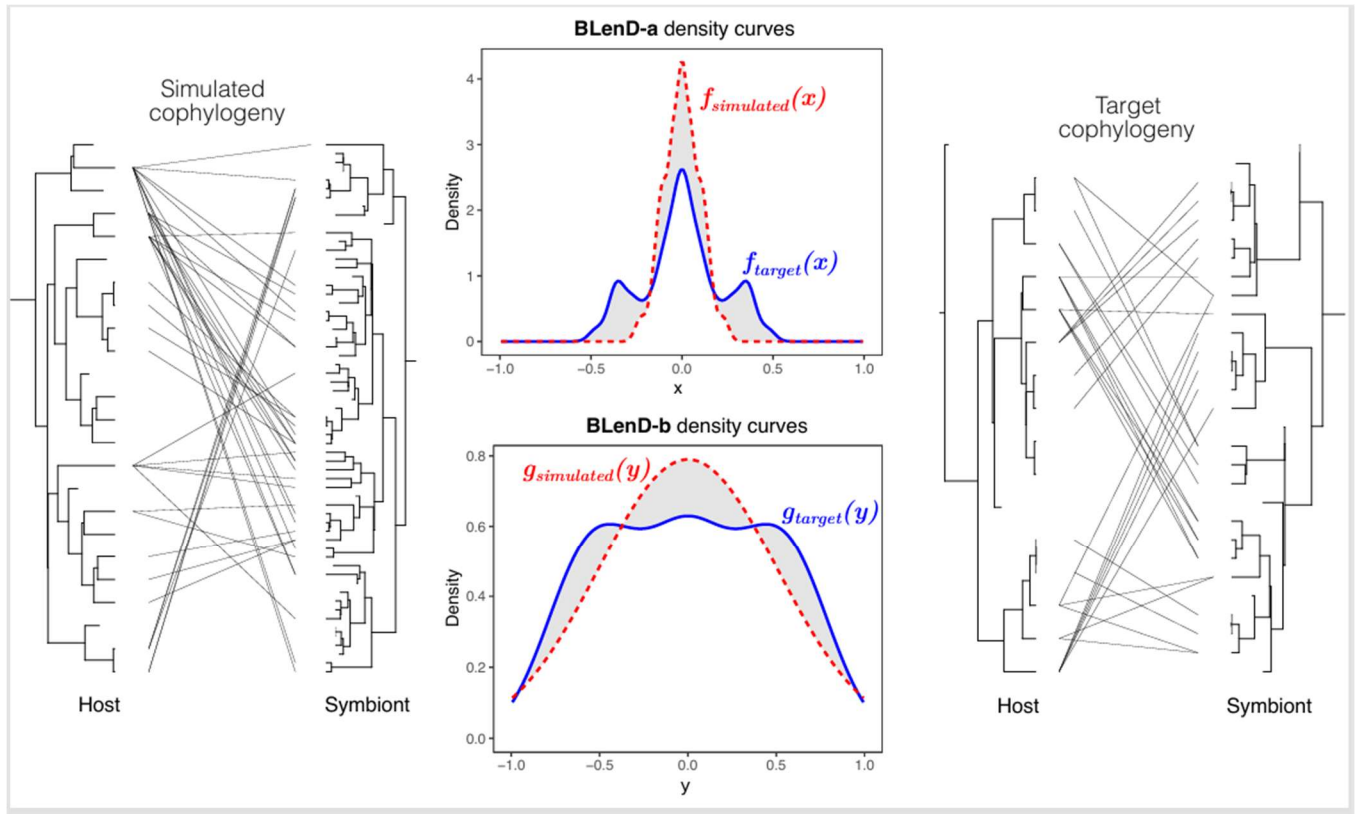


Figure 4. The distance between the target cophylogeny T and any simulated cophylogeny S based on the BLenD density curves, $dist_{BLenD-a}(S, T)$ and $dist_{BLenD-b}(S, T)$, is equal to the area between their respective BLenD curves (i.e., the shaded area). In the ABC-based parameter estimation, simulated cophylogenies with the lowest overall distance to the true cophylogeny are accepted and the parameters used to simulate the accepted cophylogenies constitute samples of the posterior distribution. Cophylogenies are shown with the extinct lineages included; however, the BLenD curves are based on extant lineages only (see Fig. 3).

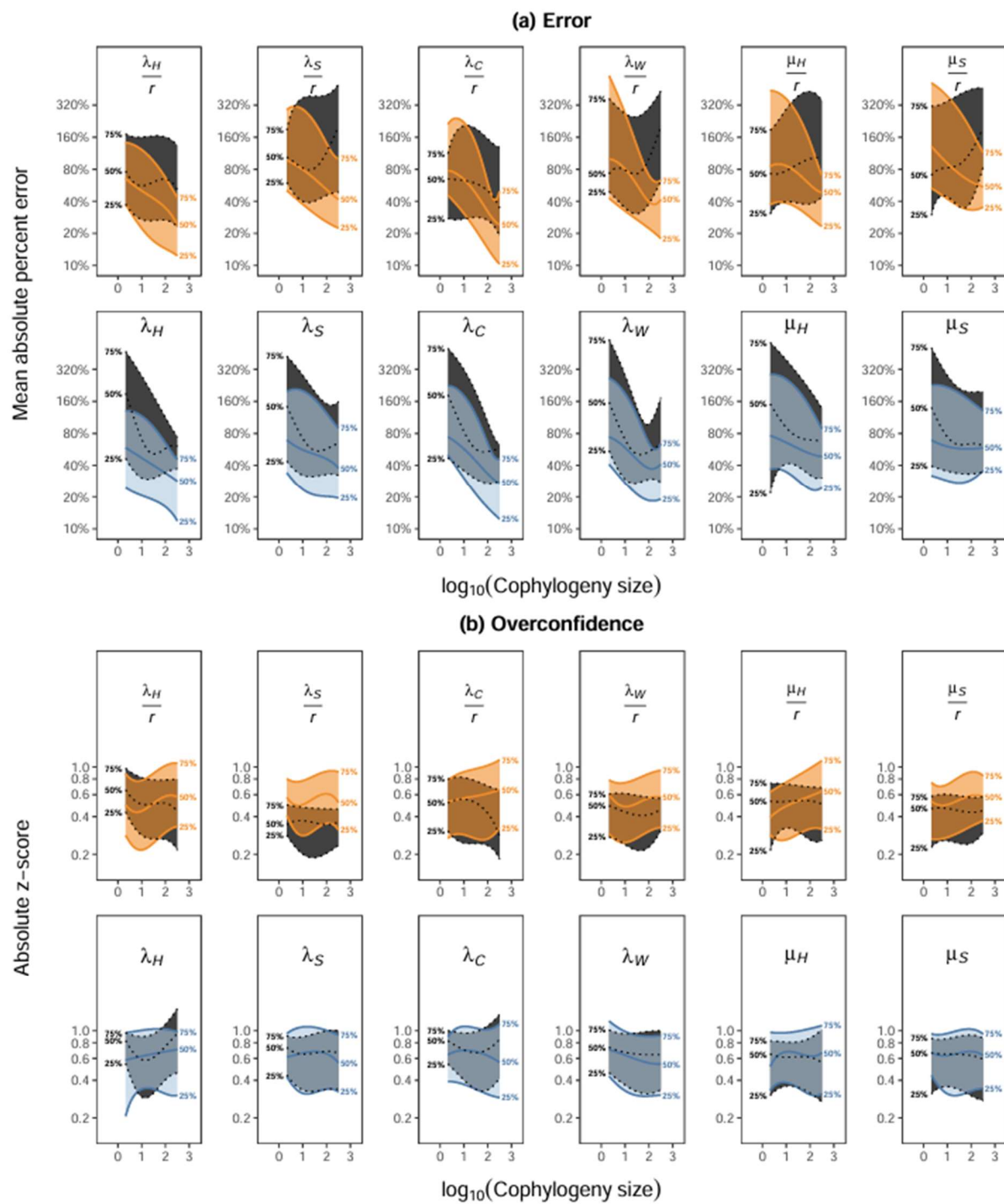


Figure 5. 25%, 50%, and 75% percentiles of the relative error and z-score of the relative and absolute rate estimates, plotted against cophylogeny size. The percentile lines are estimated based on quantile regressions of the 1000 cophylogenies in the test set (see Figs. S1-S4 for the data points used in these regressions). The dashed lines are for the prior distributions, whereas the solid lines are for the corresponding posterior distributions. A 50% quantile regression line,

for example, indicates that there is a 50% probability that the relative error (or z-score) is below the regression line.

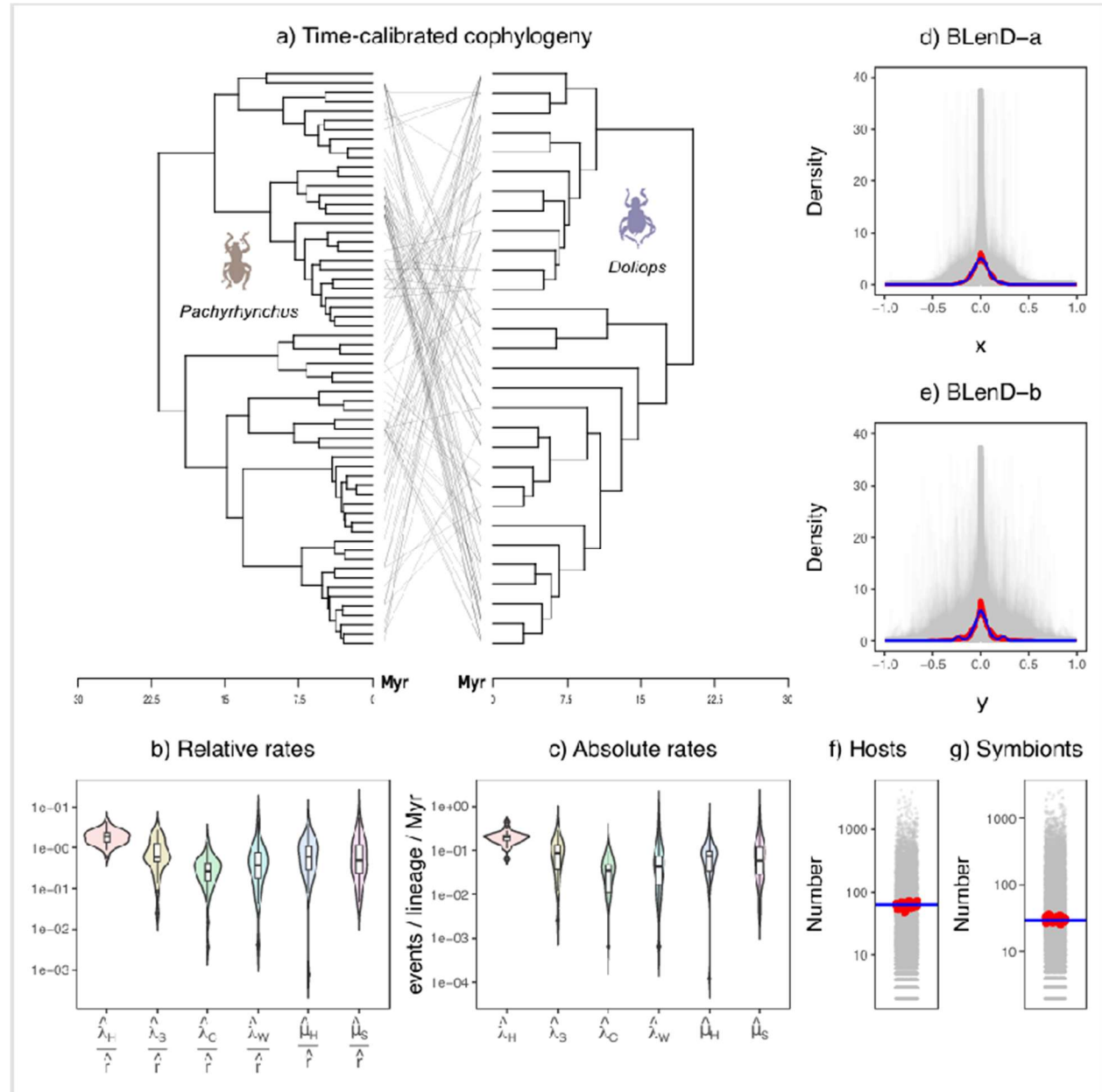


Figure 6. Inference of speciation and extinction rates from the cophylogenetic dataset of weevil mimicry (Van Dam et al. 2024): λ_H - host speciation rate; λ_S - rate of symbiont speciation without host switching; λ_C - cospeciation rate; λ_W - rate of symbiont speciation with host switching, μ_H - host extinction rate; μ_S - symbiont extinction rate; r - net diversification rate of the cophylogeny. (a) Time-calibrated cophylogeny of *Pachyrhynchus* (models, treated as hosts) and *Doliops* (mimics, treated as symbionts). (b) The posterior distribution of the relative speciation and extinction rate estimates (the time frame of the cophylogeny is not used; see Fig.

S5 for both the prior and posterior distributions). (c) The posterior distribution of the absolute speciation and extinction rate estimates (the time frame of the cophylogeny is estimated to be 27.5 Myr and used for inference; see Fig. S6 for both the prior and posterior distributions). (d, e, f & g) Posterior predictive checks of model fit. Each panel focuses on the BLenD-a curve, BLenD-b curve, host tree size, and symbiont tree size, respectively. Blue represents the real cophylogeny of beetle mimicry; red represents the simulated cophylogenies that are accepted in the ABC due to their short distance to the real (target) cophylogeny (i.e., corresponding to the posterior distribution of parameters); gray represents all the simulated cophylogenies (corresponding to the prior distribution of parameters).